

Independent Rederivation Note

A First-Principles Force-Based Derivation of Rocket Motion in a Uniform Gravitational Field

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Abstract

This note presents an independent, first-principles derivation of the rocket equation in a uniform gravitational field. Instead of using standard momentum conservation, the derivation directly analyzes the dynamics of a differential fuel mass element $-dm$ and the rocket body using Newton's second and third laws. By explicitly employing unit vectors ($\hat{i}$), enforcing a consistent sign convention across all kinematic and dynamic parameters, and properly accounting for the differential mass change ($dm < 0$), a robust physical model is constructed.

1 System Setup and Relative Kinematics

Consider a rocket launching vertically relative to a fixed inertial frame. Taking the upward direction as $+\hat{i}$, all kinematic and dynamic parameters (v, v_r, v_f, g) are defined strictly as positive scalar magnitudes ($v, v_r, v_f, g > 0$). Their physical directions are explicitly handled by unit vectors ($\hat{i}$ and $-\hat{i}$), ensuring that direct substitution of magnitudes into the final equation requires no further sign adjustments:

- v : Upward speed of the rocket relative to the ground ($\vec{v} = v\hat{i}$)
- v_f : Downward speed of the ejected fuel relative to the ground ($\vec{v}_f = -v_f\hat{i}$)
- v_r : Exhaust speed relative to the rocket ($\vec{v}_r = -v_r\hat{i}$)
- g : Acceleration due to gravity ($\vec{g} = -g\hat{i}$)

From the definition of relative velocity:

$$\begin{aligned}\vec{v}_r &= \vec{v}_f - \vec{v} \\ \implies -v_r\hat{i} &= -v_f\hat{i} - v\hat{i} \\ \implies v_r &= v_f + v \\ \therefore v_f &= v_r - v\end{aligned}\tag{1}$$

2 Dynamics of the Ejected Fuel Element ($-dm$)

Let m denote the total mass of the rocket system (including remaining fuel) at time t . Over an infinitesimal time duration dt , the mass of the rocket system changes from an initial value m'_i

to a final value m'_f . Since fuel is continuously expelled, $m'_f < m'_i$, which yields an inherently negative differential change in rocket mass:

$$dm = m'_f - m'_i < 0$$

Since physical mass cannot be negative, the mass of the fuel element ejected during dt is equal to the mass lost by the rocket system, given by $-dm > 0$.

Prior to ejection, the fuel element moves alongside the rocket body with an initial velocity equal to that of the rocket, $\vec{v} = v\hat{i}$, relative to the ground. Over the time interval dt , as it is expelled from the rocket, its velocity relative to the ground changes to $\vec{v}_f = -v_f\hat{i}$. The average acceleration $\vec{a}_f$ of the ejected fuel element during dt which acts in the downward direction is therefore:

$$\vec{a}_f = \frac{\vec{v}_f - \vec{v}}{dt} = \frac{-v_f\hat{i} - v\hat{i}}{dt} = \frac{-(v_f + v)\hat{i}}{dt} = -\frac{v_r}{dt}\hat{i} \quad (2)$$

The forces acting on the fuel element $-dm$ are:

1. Gravitational force (Acts in the downward direction): $\vec{W}_f = -[(-dm)g]\hat{i} = (dm)g\hat{i}$
2. Downward interaction force exerted by the rocket body on the ejected fuel: $\vec{F}_f = -F_f\hat{i}$

Applying Newton's Second Law ($\sum \vec{F} = (-dm)\vec{a}_f$):

$$\begin{aligned} \vec{F}_f + \vec{W}_f &= (-dm)\vec{a}_f \\ \implies -F_f\hat{i} + (dm)g\hat{i} &= (-dm)\left(\frac{-v_r}{dt}\right)\hat{i} \\ \implies -F_f + (dm)g &= (dm)\frac{v_r}{dt} \\ \therefore F_f &= v_r\left(-\frac{dm}{dt}\right) + (dm)g \end{aligned} \quad (3)$$

3 Dynamics of the Main Rocket Body ($m + dm$)

The instantaneous mass of the main rocket body after the ejection of the fuel element is

$$m - (-dm) = m + dm,$$

where m is the rocket system mass immediately before the infinitesimal mass ejection and $-dm > 0$ is the mass expelled during dt .

The total forces acting on the rocket body are the reaction force $\vec{F}_R = F_R\hat{i}$ exerted by the ejected fuel in the upward direction and its weight $\vec{W}_R = -(m + dm)g\hat{i}$ in the downward direction. The rocket acceleration that acts in the upward direction is:

$$\vec{a}_R = \frac{(v + dv)\hat{i} - v\hat{i}}{dt} = \frac{dv}{dt}\hat{i} \quad (4)$$

Newton's Second Law gives:

$$\begin{aligned} \vec{F}_R + \vec{W}_R &= (m + dm)a_R\hat{i} \\ \implies F_R\hat{i} - (m + dm)g\hat{i} &= (m + dm)\frac{dv}{dt}\hat{i} \\ \therefore F_R &= (m + dm)g + (m + dm)\frac{dv}{dt} \end{aligned} \quad (5)$$

4 Force Balance and Equation of Motion

By Newton's Third Law, the force exerted by the ejected fuel on the rocket body is equal in magnitude and opposite in direction to the force exerted by the rocket body on the ejected fuel:

$$\begin{aligned}\vec{F}_R &= -\vec{F}_f \\ \implies F_R \hat{i} &= -(-F_f \hat{i}) \\ \therefore F_R &= F_f\end{aligned}\tag{6}$$

Putting Equations (3) and (5) in Equation (6):

$$v_r \left(-\frac{dm}{dt} \right) + (dm)g = (m + dm)g + (m + dm) \frac{dv}{dt}$$

Expanding the right hand side:

$$v_r \left(-\frac{dm}{dt} \right) + (dm)g = mg + (dm)g + m \frac{dv}{dt} + (dm) \frac{dv}{dt}$$

Canceling $(dm)g$ from both sides and neglecting the higher-order differential product term $(dm) \frac{dv}{dt} \approx 0$:

$$\begin{aligned}v_r \left(-\frac{dm}{dt} \right) &= mg + m \frac{dv}{dt} \\ \therefore m \frac{dv}{dt} &= v_r \left(-\frac{dm}{dt} \right) - mg\end{aligned}\tag{7}$$

which is the differential equation of rocket motion in an uniform gravitational field.

5 Acceleration and Integration

Rearranging Equation (7) to express rocket acceleration $a_R = \frac{dv}{dt}$:

$$a_R = \frac{dv}{dt} = \frac{v_r}{m} \left(-\frac{dm}{dt} \right) - g\tag{8}$$

Here, $\left(-\frac{dm}{dt} \right) > 0$ represents the positive rate of fuel ejection. Expressing in differential form:

$$dv = -v_r \frac{dm}{m} - gdt\tag{9}$$

For constant exhaust speed v_r and uniform gravitational acceleration g , integrating from initial state $(t = 0, v = 0, m = m_i)$ to final state $(t = t, v = v, m = m_f)$:

$$\begin{aligned}\int_0^v dv &= -v_r \int_{m_i}^{m_f} \frac{dm}{m} - g \int_0^t dt \\ \implies [v]_0^v &= -v_r [\ln m]_{m_i}^{m_f} - g[t]_0^t \\ \implies v - 0 &= -v_r (\ln m_f - \ln m_i) - g(t - 0) \\ \implies v &= -v_r \ln \left(\frac{m_f}{m_i} \right) - gt \\ \therefore v &= v_r \ln \left(\frac{m_i}{m_f} \right) - gt\end{aligned}\tag{10}$$

6 Conclusion

This note presented a direct, force-based derivation of the rocket equation in a uniform gravitational field. By applying Newton's second and third laws directly to both the ejected fuel element and the rocket body, the governing equations were derived without relying on system-wide momentum conservation. The systematic use of unit vectors and explicit tracking of the negative mass differential ($dm < 0$) keep the sign conventions consistent throughout the derivation, allowing parameter magnitudes to be substituted directly into the final equation. Ultimately, this approach offers a clear physical perspective on how reaction forces drive rocket motion, providing an intuitive reference for analyzing variable-mass systems.